MEASUREMENTS, SIGNIFICANT FIGURES, AND ERRORS

The history of systems of measurement can be traced to the first recorded civilizations in the world until the systems were made and established to be more cohesive in the 18th century. Ancient people would use their foot, hand, arm to measure lengths of objects around until the eventual inventions of devices such as rulers, calipers, and others through the centuries.

To formally establish the universal system of measurements, in 1960 the Systeme International d'Unites or International System of Units (SI), commonly known as the metric system, was established by the 11th General Conference of Weights and Measures. The "SI unit" for length is meter; gram for mass; liter for volume/capacity; and Newton for force, to mention a few. The SI uses the prefixes for powers of 10 for length, volume, and mass: a kilogram means 1000 grams; a milliliter (mL) means 10^-3 of a liter; and a centimeter for 10^-2 of a meter.

On the other hand, the Imperial or the English system uses foot (length), inch (length), pound (mass and force), and gallon (capacity). These two systems are widely used depending on the nature of the objects being measured.

Alongside technological advancement, scientists have established the concepts of accuracy and precision across lab experiments to ensure any significant deviation would be avoided to keep the data exact or at least approaching the true value. A certain device could measure 100 cm but another could render 100.04 cm, but the true value is 101 cm. Such a scenario in the lab is inevitable but can be prevented with the correct use of appropriate tools, and further knowledge on basic measurement.

Some tools used in measuring small lengths or distances are vernier calipers and micrometers.

Micrometer Caliper

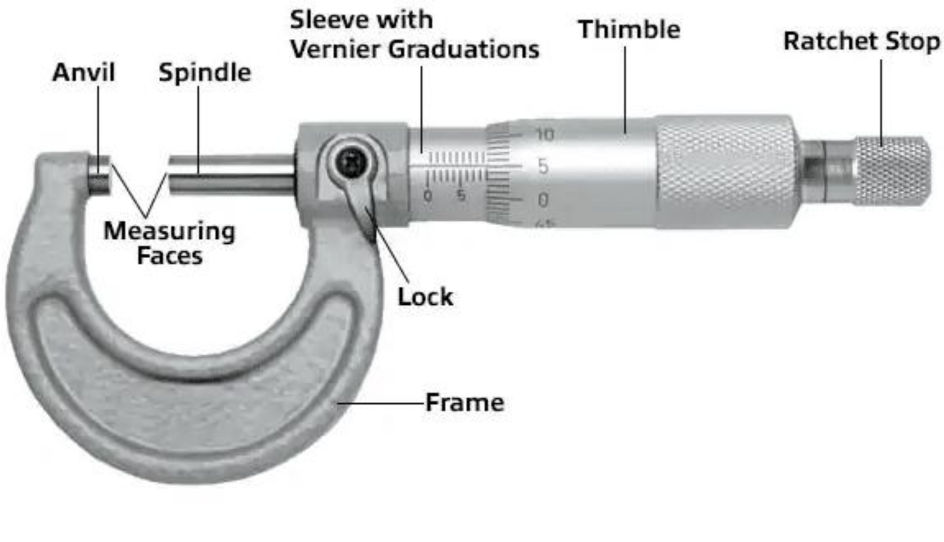

Figure 1.1. Parts of a Micrometer.

The micrometer is used to measure solid objects accurately that may fit its spindle and anvil. It is calibrated using the knob. It gives a reading in millimeters. Inventor: William Gascoigne (1612-1644), England.

Vernier Caliper

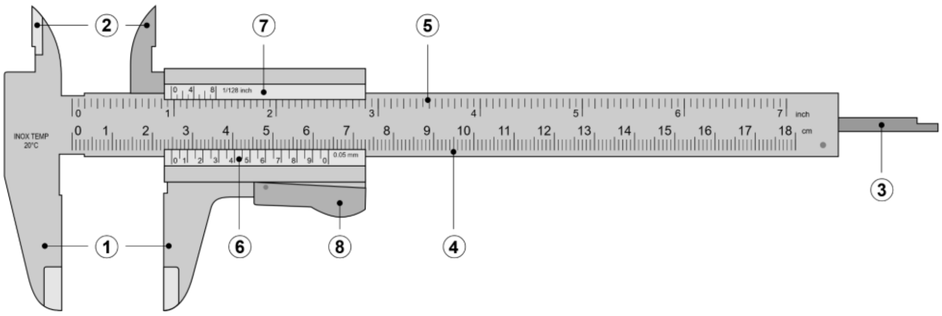

Figure 1.2. Parts of a Vernier Caliper.

Legend:

1. Lower Jaw - measures outer dimension
2. Upper Jaw - measures inner dimension
3. Blade - measures depth
4. Main Scale (in centimeter unit) - the fixed scale
5. Main Scale (in inch unit) - the fixed scale
6. Vernier Scale (centimeter unit) - the sliding scale or the movable scale
7. Vernier Scale (inch unit) - the sliding scale or the movable scale
8. Knurled Wheel - sliding jaw

The vernier caliper is best used for measuring hollow right circular cylinders as it can measure outer/inner dimensions, and depth. Inventor: Pierre Vernier (1584-1638), France.

SIGNIFICANT FORMULAS

Mean/Average

$$Mean = \frac{\Sigma measurements}{number\ of\ measurements}$$

Volume of Cylinder

$$V_{cylinder} = \frac{\pi({MeanDiameter_{out}}^2 - {MeanDiameter_{in}}^2)}{4} \times MeanHeight$$

Experimental Density

$$\rho = \frac{mass\ of\ the\ object}{volume\ of\ the\ object} = \frac{m}{V}$$

Volume of a Sphere

$$V_{sphere} = \frac{\pi}{6} \times MeanDiameter^3$$

Percentage Error

$$\%Error = \left|\frac{\rho_{theo} - \rho_{exp}}{\rho_{theo}}\right| \times 100$$